Jyothish J

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.14
Bhavans Vidya Mandir, Chithali, Palakkad	Class XII (CBSE)	2023	96.20%
Bhavans Vidya Mandir, Chithali, Palakkad	Class X (CBSE)	2021	95.20%

SCHOLASTIC ACHIEVEMENTS

Best Out Going Student Award - Bhavans Vidya Mandir Chithali

- Recognized for outstanding academic performance, leadership, character, attitude, and overall contribution to the school community.

SKILLS

- **Programming Languages:** Python, C, C++, MATLAB, TypeScript, JavaScript
- **Web Development:** React, HTML5, CSS3, FastAPI, Supabase
- **Tools & Databases:** Git, GitHub, PostgreSQL
- **Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, System Design

PROJECTS

ProductivityTube - AI-Powered Educational Video Platform

- **Live:** https://productivity-tube.vercel.app
- Developed a full-stack educational video platform using **React, Python, and YouTube Data API** that filters YouTube content to provide a distraction-free learning experience.
- Designed a **dual-LLM pipeline** using **Groq** for low-latency filtering and **Gemini** for background processing, balancing response time with API rate limits.
- Implemented **database caching** for previously processed searches and a **sliding-window recommendation engine** to deliver personalized suggestions while minimizing storage requirements.
- Engineered a secure **BYOK (Bring Your Own Key)** architecture with **AES-encrypted API key storage**, **Google OAuth**, **Row-Level Security (RLS)**, and optimized backend performance using **asynchronous Python**.

Generalized Planar Mechanism Solver

- **Live:** https://kinematic-engine.vercel.app/
- Developed an **interactive web application** that enables users to draw and simulate arbitrary planar mechanisms with an unrestricted number of links and sliders.
- Designed a **generalized object-oriented kinematic solver** using the **Newton-Raphson method** with a **numerical Jacobian**, eliminating the need for mechanism-specific analytical equations.
- Validated the framework on mechanisms with **50+ links**, demonstrating scalability and extensibility across complex linkage systems.
- Currently extending the solver to support **velocity, acceleration, and dynamic analysis** using Jacobian-based formulations.

Programmable 8-Bit Computer

- Designed and built a programmable SAP-1 inspired 8-bit computer using discrete TTL logic.
- Engineered a microcode-based control unit capable of executing arithmetic, memory, and branching instructions.
- Extended the system with an Arduino-based interpreter for simplified program loading and execution.

LearnSpace - AI-Powered Learning Platform | Live: Link

- Designed an integrated learning workspace to reduce context switching by combining notes, tasks, AI assistance, and cloud resources.
- Architected a scalable BYOK model with Google Drive integration, lowering infrastructure costs while preserving user ownership of data.
- Implemented secure multi-tenant data management using AES encryption and PostgreSQL Row-Level Security.

KnowYourCase - AI-Powered Legal Assistance Platform *(In Progress)* *prototype link*

- Designed the MVP and interactive prototype for a platform connecting users with lawyers for quick legal guidance.
- Defined the product architecture for a two-sided marketplace featuring lawyer profiles, AI-assisted consultations, and future AI-based legal skill assessment.
- Currently developing the backend, authentication, and consultation workflows.

POSITIONS OF RESPONSIBILITY

Technical Head Cepheid- Astronomy and Astrophysics Club

- Led a team of **50+ core members** in planning and executing astronomy outreach, observation sessions, and club events.
- Mentored members on technical projects and guided the development of astronomy and aerospace related initiatives.

Web Development Coordinator - Alumni Cell, IIT Hyderabad

- Collaborated with the team to the design, development, and maintenance of the Alumni Cell website.

Core Member, Aeolus - Motorsports Club, IIT Hyderabad

- Contributed to the design and development of a Formula Student race car as part of the **Vehicle Dynamics** team.
- Performed sensitivity analysis using **Optimum Lab** and analyzed tire data to optimize **camber, toe, and roll characteristics**; currently working on **FEA-based component optimization**.

Core Member Aero Club, IIT Hyderabad